# Joint vertex covers and the envelope of Gowers's inverse induction

Formalization notes, Combinatorics/Ramsey (2026-10-11)

**Abstract**

In Gowers's proof of Szemerédi's theorem, the structural step of Section 16 (the single-piece lift) must produce one multilinear map that agrees with the *remainder* $\varphi''$ on a dense piece of a box. The remainder is a signed sum over the $2^k-1$ non-top vertices of the cube $\{0,1\}^k$, and each vertex contributes a function that lives on a lower-dimensional face. If the vertices are handled one after another, each at the density left by the previous one, the degrees of the density losses multiply, and the resulting envelope grows like $3^k$ in the doubly-logarithmic scale. That outgrows the printed budgets $2^{d+10}$ (Theorem 18.1) and $2^{L+9}$ (Theorem 18.2) at degree 10 and length 14. We show that the vertex *covers* can be composed at a precision fixed in advance, with a single pigeonhole at the end. The losses then add, the envelope grows like $2^k$, and Theorems 18.1, 18.2 and Corollary 18.7 hold in every degree and length, at the printed constants and with some room to spare. Several of the printed bounds turn out to be comparison artifacts: the Fejér selection in Lemma 15.5, exact class degrees in small dimensions, and polynomial bounds for the Bohr radius, the base budget and Lemma 15.6. Removing them gives Theorem 18.1 with $d+5$ in every degree, $d+2$ from degree 15; and Theorem 18.2 with $L+3$ in every length, and with no offset at all from length 18. Every statement below is formalized in Lean (module names are given in §9).

## 1 Covers

Throughout, $N$ is prime, a *box* is a product of arithmetic progressions in $\mathbb{Z}_N$ with a common difference, and $w(P)$ is the least side length of a box $P$. A map is *multilinear* in the sense of the paper.

**Definition 1** (Cover)**.** Let $\Gamma \subseteq \mathbb{Z}_N^d \times \mathbb{Z}_N$ be a relation and $Q, E\colon (0,1] \to \mathbb{R}$. We say $\Gamma$ is *covered with controls* $(Q,E)$ if for every $\theta \in (0,1]$ and every proper box $P$ there are

- a partition of $P$ into proper boxes $Q_1, \dots, Q_M$ with $w(Q_j) \geq w(P)^{E(\theta)}$,
- a set $H \subseteq P$ with $|H| \geq (1-\theta)|P|$,
- for each $j$ at most $Q(\theta)$ multilinear maps $\mu_{j,1}, \dots, \mu_{j,q}$,

such that every $(x,y) \in \Gamma$ with $x \in Q_j \cap H$ satisfies $y = \mu_{j,i}(x)$ for some $i$.

The precision $\theta$ is chosen by the user of the cover *before* it is applied, and the loss $\theta|P|$ is additive. A *provider*, by contrast, takes a set of density $t$ and returns a piece of density $c(t)$; composing providers composes the functions $c$.

## 2 Joint covers

**Lemma 2** (Joint cover)**.** *Let $E$ be a finite index set, and for $e \in E$ let $f_e$ be a function on a domain $D_e \subseteq \mathbb{Z}_N^d$ whose graph is covered with controls $(Q_e, E_e)$, with $Q_e, E_e \geq 0$. Fix precisions $\theta_e \in (0,1]$.*

*Then every proper box $P$ has a partition into proper boxes $R_1, \dots, R_M$, a set $G \subseteq P$ and, for each $j$, at most $q \le \prod_e Q_e(\theta_e)$ tuples $(\nu_{j,i,e})_{e\in E}$ of multilinear maps, such that*

$$|G| \ge \Big(1 - \sum_e \theta_e\Big)|P|, \qquad w(R_j) \ge w(P)^{\prod_e E_e(\theta_e)},$$

*and every $x \in R_j \cap G$ lying in all domains $D_e$ has an index $i$ with $f_e(x) = \nu_{j,i,e}(x)$ for all $e$.*

*Proof.* Induction on $E$. For $E = \emptyset$ take the trivial partition, $G = P$ and one empty tuple. For $E \cup \{a\}$, apply the induction hypothesis to $P$, obtaining cells $Q_j$, a good set $G$ and tuples $\nu_{j,i}$. Apply the cover of $f_a$ at precision $\theta_a$ to every cell $Q_j$; this gives sub-cells $R_{j,l}$, good sets whose union $G'$ satisfies $|G'| \ge (1-\theta_a)|P|$ (the cells partition $P$), and maps $\mu_{j,l,i'}$. Take the sub-cells $R_{j,l}$ as the new partition, the good set $G \cap G'$, of size at least $|G| + |G'| - |P| \ge (1 - \sum_E \theta_e - \theta_a)|P|$, and the product tuples $(\nu_{j,i}, \mu_{j,l,i'})$. The width bound follows from $w(R_{j,l}) \ge w(Q_j)^{E_a} \ge w(P)^{E_a \prod_E E_e}$. □

**Corollary 3** (One piece)**.** *In the setting of Lemma 2, put $s = \sum_e \theta_e$ and $Q = \prod_e Q_e(\theta_e)$. If $H \subseteq P$ lies in every domain $D_e$ and $|H| \ge \theta|P|$ with $\theta > s$, then some cell $R$ and some maps $\nu_e$ satisfy*

$$\big|\{x \in H \cap R : f_e(x) = \nu_e(x) \text{ for all } e\}\big| \ge \frac{\theta - s}{Q}\,|R|.$$

*Proof.* $|H \cap G| \ge (\theta - s)|P|$, so some cell $R$ has $|H \cap G \cap R| \ge (\theta - s)|R|$. Each of its points is covered by one of at most $Q$ tuples; the most popular tuple covers a $1/Q$ fraction. □

The only density loss is the single factor $1/Q$, and $Q$ is a product of counts evaluated at *fixed* precisions. If $Q_e(\theta) \le (C/\theta)^{D_e}$, the density has degree $1 + \sum_e D_e$ in $1/\theta$: the degrees add.

## 3 The remainder of the single-piece step

On the good domain of the step, the remainder is $\varphi''(x) = -\sum_{e\neq \mathbf{1}} (-1)^{k+|e|}\varphi_e(x)$, where $\varphi_e$ is the translated cube vertex $e$, a pullback of $\varphi$ to the face spanned by the directions $S_e$, of dimension $|e|+1$. By induction on the dimension, the face carries a cover of degree $D_{|S_e|}$. It lifts to the ambient box without any loss in the precision: the count becomes $\max(Q, 3^{k+1})$ and the exponent $E/16^{k+1-|S_e|}$. Applying Corollary 3 with $\theta_e = \theta_2/2^{k+1}$ to the dense good set (density $\theta_2$) gives one map $M'' = -\sum_e (-1)^{k+|e|}\nu_e$ on a piece of density

$$\rho_1 = \frac{\theta_2/2}{\prod_{e\neq\mathbf{1}} \max\big(Q_{|S_e|}(\theta_2/2^{k+1}), 3^{k+1}\big)}$$

and width at least $N^{\prod_e E_e/16^{k+1-|S_e|}}$.

*Remark* 4 (Nesting versus joint covers). With the class degrees $D_l = 2^{68\cdot 2^l}$, nesting one provider per vertex gives $\log_2 \deg \rho_1^{-1} \approx \sum_e 68 \cdot 2^{|e|+1} = 136 \cdot 3^k$. The joint cover gives $\log_2\big(1 + \sum_e D_{|S_e|}\big) \le 68 \cdot 2^k + k$.

## 4 The envelope

Write $X = 2/\alpha$. The inverse induction carries bounds $\beta^{-1}, \sigma^{-1} \le X^{2^\mu}$ from degree $d-1$ to degree $d$. With the joint-cover remainder, one step in dimension $k$ costs

$$\mu_{\text{new}} = \mu + \underbrace{(68 \cdot 2^k + \mu_0 + k + 3)}_{\rho_1} + 2 \cdot 68 \cdot 2^k + O(k^2),$$

where $\mu_0$ bounds the base budget of the cover controls. With the standard $\mu_0 \approx 64 \cdot 2^k$ one step costs $268 \cdot 2^k + O(k^2)$. Hence

**Theorem 5.** *For every $k \geq 1$ the Theorem 18.1 envelope satisfies $\mu_k \leq 1000 \cdot 2^k$, and the Theorem 18.2 envelope (which also carries thresholds) satisfies $\mu_k \leq 1000 \cdot 2^k$ as well. Both are well inside the printed budgets $2^{d+10} = 4096 \cdot 2^{d-2}$ and $2^{L+9} = 8192 \cdot 2^{L-4}$. Consequently Theorems 18.1 and 18.2 and Corollary 18.7 hold in every degree and length.*

## 5 Sharper constants

The same proofs give smaller tower constants. Moreover, the class base $V = 1 + \sum_{j<k}(K_j + p_j)$ satisfies $V \leq 2^{2^{2^{k+4}+1}}$, so the base budget $\mu_0 = 2^{k+5} + 9$ suffices; one step then costs $236 \cdot 2^k + O(k^2)$ and the envelopes grow like $472 \cdot 2^k < 512 \cdot 2^k$.

| | printed | standard base | tight base | Fejér $\theta_1$ (§6) |
|---|---|---|---|---|
| Theorem 18.1, exponent $\alpha^{2^{2^{d+c}}}$ | $c = 10$ | $c = 8$ $(d \geq 4)$ | $c = 7$ | $c = 6$ (every $d$) |
| Theorem 18.2, threshold with $L + c$ | $c = 9$ | $c = 6$ (every $L$) | $c = 5$ | $c = 4$ (every $L$) |
| Corollary 18.7, threshold with $L + c$ | $c = 9$ | $c = 6$ (every $L$) | $c = 5$ | $c = 4$ (every $L$) |
| $W(k, r) \leq 2^{2^{r^{2^{2^{k+c}}}}}$, $r \geq 2$ | $c = 9$ | — | $c = 5$ | $c = 4$ |
| Theorem 1.3, density $(\log\log N)^{-2^{-2^{k+c}}}$ | $c = 9$ | — | $c = 5$ | $c = 4$ |

The $W(k, r)$ row is Theorem 18.2 at density $\delta = 1/r$: in an $r$-colouring of $[1, N]$ one colour class has at least $N/r$ elements. This gives Gowers's van der Waerden bound, sharpened from $k + 9$ to $k + 5$. The headline Theorem 1.3 follows from Theorem 18.2 by a calculation in which the constant enters only through $2^{k+9}$; repeated with $k + 5$, it gives the density exponent $2^{-2^{k+5}}$.

With the printed $\theta_1$ the method stops at the tight-base constants. The envelope recursion grows like $472 \cdot 2^k$, and $c = 6$ for Theorem 18.1 (or $c = 4$ for Theorem 18.2) needs a step cost below $128 \cdot 2^k$, while the vertex covers and the two slice controls alone cost $204 \cdot 2^k$. The last column of the table comes from changing $\theta_1$ itself (§6).

For Theorem 18.2 at $c = 5$ one more point matters. The modulus threshold of the inverse step contains two pure constants, the Section 5 refinement constant $\leq 2^{2^{40d^3+2}}$ and a boundary constant $\leq 2^{2^{1083}}$. A cruder bound charges them at the level of powers of $X$, which puts a cubic term $40(k+2)^3$ into the envelope and blocks $L + 5$ for $L \leq 10$. The threshold is only needed below $\exp(\exp(X^{2^\mu}))$, and $2^{2^m} \leq \exp(X^{2^j})$ as soon as $m \leq 2^j$. At that level the constants cost only $k + 16$, and $L + 5$ holds in every length.

## 6 A Fejér selection and smaller constants

The class degree $68 \cdot 2^l$ and the base budget both come from the Section 16 density $\theta_1 = (\theta\gamma/4)^{2^{2^{k+5}}}$. Its double exponential is created in Lemma 15.5. A point is kept with probability $(1+\cos)/2$, which gives a respected arrangement of $M = 2^{k+4}$ vertices an advantage of only $1 + 2^{1-M}$ per stage, so $2^{M-1}$ stages are needed.

*One Fejér stage.* Keep a point with the Fejér weight $F_m(u) = |\sum_{j \leq m} e(ju/N)|^2/(m+1)^2$ of order $m + 1 = M\lceil 6/(\alpha\eta)\rceil$. Averaged over the coefficients, the survival probability of an arrangement is a sum over the integer vectors $\eta \in [-m, m]^M$ annihilating all its monomials. For arrangements that are not $m$-degenerate, these are the multiples $j\varepsilon$ of the parity vector. They

contribute $\sum_{|j|\le m}(1-|j|/(m+1))^M \ge T$ for respected arrangements and only $j=0$ otherwise. Lemma 15.4 extends to coefficients in $[-m,m]$, since Corollary 15.2 holds for every coefficient vector, and it bounds the $m$-degenerate arrangements. The resulting Lemma 15.5 has exponent $2^{k+4}(k+6)$, and Lemma 15.6 has

$$\theta_1' = (\theta\gamma/4)^F, \qquad F \le 2^{5k+14},$$

in place of $2^{2^{k+5}}$.

*The tight radius.* The Bohr radius $\zeta = 2^{-s}$ only has to satisfy Lemma 16.5's condition, which needs $s \ge 2^{83}/(\theta_1/4)^{11}$. With $\theta_1'$ this is $s' \le (2/\theta\gamma)^{22F+105}$, instead of the printed $(2/\theta\gamma)^{2^{2^{k+6}}}$.

*Propagation.* With these inputs the dimension-two class has degree $2^{46}$ (printed $2^{132}$). The split-class recursion keeps $\log_2 D_l \le 22\cdot 2^l$, so the class covers have degree $2^{23\cdot 2^l}$ in place of $2^{68\cdot 2^l}$. Since $\theta_1' \ge \theta_1$, the budget, $\theta_2$, $\delta$ and the radius of Section 18 are at least the printed ones, and every printed fixed-parameter bound transfers unchanged. One step of the inverse induction then costs

$$\mu_{\text{new}} = \mu + \underbrace{23\cdot 2^k}_{\text{vertices}} + \underbrace{32\cdot 2^k}_{\mu_0} + \underbrace{2\cdot 23\cdot 2^k}_{\text{slices}} + O(k^2) = \mu + 101\cdot 2^k + O(k^2),$$

and both envelopes are at most $230\cdot 2^k$ for $k\ge 4$, below the $256\cdot 2^k$ of $c=6$ and $c=4$.

**Theorem 6.** *Theorem 18.1 holds in every degree $d$ with exponent $\alpha^{2^{2^{d+6}}}$, and Theorem 18.2 and Corollary 18.7 hold in every length $L$ with $L+4$ in the threshold. Hence $W(k,r) \le 2^{2^{r^{2^{2^{k+4}}}}}$, and Theorem 1.3 holds with density exponent $2^{-2^{k+4}}$.*

*Large degrees.* The proved class bound $23\cdot 2^l$ is loose. The class recursion only needs slack that grows additively, and indeed

$$2^{l+6}a_l \le 2^{14\cdot 2^l}, \qquad D_l \le 2^{14\cdot 2^l + 3(l+4)^2}$$

in every dimension $l\ge 2$, since $3(l+5)^2 - 3(l+4)^2 = 6l+27$ covers the $5l+22$ that one step adds. Moreover the Fejér $\theta_1$ no longer forces the base budget $32\cdot 2^k$: the class-base term $16\cdot 2^k$ suffices. One step then costs $58\cdot 2^k + O(k^2)$, and the envelopes stay below $128\cdot 2^k$ from $k=10$ on.

**Theorem 7.** *Theorem 18.1 holds with exponent $\alpha^{2^{2^{d+5}}}$ at every degree $d\ge 12$, and Theorem 18.2 and Corollary 18.7 hold with $L+3$ at every length $L\ge 14$; in particular $W(k,r) \le 2^{2^{r^{2^{2^{k+3}}}}}$ for $k\ge 14$, and Theorem 1.3 holds with density exponent $2^{-2^{k+3}}$ for $k\ge 14$.*

The small cases stay at $d+6$ and $L+4$ because of the class recursion's transient near $l=4$, where the family exponent squares, $a_{l+1} \approx 2^{l+5}a_l^2$.

*The dimension-two family exponent.* Since the family exponent squares, $\log_2 a_l \approx (\log_2 a_2/4)\,2^l$, so the constant 14 above is set by $a_2 = 2^{46}$. That value only reflects the passage from a power class, which sets the family exponent equal to the parameter degree. The dimension-two controls are polynomials of fixed degree in the number $M = n\,m$ of pieces: the base controls are linear in $M$. In the two bases $\nu = n+1$ and $W = 2V/(\gamma\theta\rho)$:

- the count is at most $\nu^{11}W^{2^{41}}$;
- the inverse width exponent is at most $\nu^{29}W^{2^{46}}$.

So dimension two is a split class with $a_2 = 32$ and $D_2 = 2^{48}$, and the invariants become

$$2^{l+6} a_l \le 2^{4 \cdot 2^l}, \qquad D_l \le 2^{4 \cdot 2^l + 3(l+4)^2 + 30}.$$

The class degree is then $4 \cdot 2^l + 3(l+4)^2 + 32$.

At this class degree the density degree of the step is about $28 \cdot 2^k$. That is below the $32 \cdot 2^k$ which the $\eta$ step of the printed envelope asks of it, but the $\eta$ step only absorbs the constant $2^{2(k+2)^3}\, 2^{k+1}(2/\alpha)^2$, and $3(k+4)^2$ suffices for that. One step then costs $28 \cdot 2^k + O(k^2)$, so the Theorem 18.1 envelope tends to $56 \cdot 2^k < 64 \cdot 2^k$.

**Theorem 8.** *Theorem 18.1 holds with exponent $\alpha^{2^{2^{d+4}}}$ at every degree $d \ge 13$, and with $\alpha^{2^{2^{d+5}}}$ at every degree $d \ge 9$. Theorem 18.2 and Corollary 18.7 hold with $L+3$ at every length $L \ge 11$; in particular $W(k,r) \le 2^{2^{r^{2^{2^{k+3}}}}}$ and Theorem 1.3 holds with density exponent $2^{-2^{k+3}}$ for $k \ge 11$.*

This Theorem 18.2 envelope cannot reach $L+2$: its threshold branch alone is $64 \cdot 2^k + 8 > 64 \cdot 2^k$ (but see Theorem 10). Nor can $d+3$ be reached, since the base budget contributes $16 \cdot 2^k$ to every step.

*Exact small dimensions.* In small dimensions the slack $3(l+4)^2 + 32$ dominates the class degree, but there the split degrees are explicit numbers.

| $l$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| true | 17 | 49 | 85 | 136 | 210 | 326 | 524 | 881 | 1550 | 2841 | 5371 | 10375 |
| bound | 115 | 156 | 211 | 288 | 403 | 588 | 907 | 1488 | 2587 | 4716 | 8899 | 17184 |

The table compares their log-degrees $\lceil \log_2(a_l + D_l + l + 1) \rceil$ with the bound $4 \cdot 2^l + 3(l+4)^2 + 32$; the kernel evaluates them. With these degrees the envelope peaks at $124.4 \cdot 2^k$ (at $k = 3$), below $128 \cdot 2^k$. The kernel also evaluates the envelope up to $k = 12$, and the invariants carry it beyond. Degrees $d \le 3$ and lengths $L \le 5$ come from the explicit corpus bounds.

**Theorem 9.** *Theorem 18.1 holds with exponent $\alpha^{2^{2^{d+5}}}$ in every degree $d$, and with $\alpha^{2^{2^{d+4}}}$ at every degree $d \ge 11$. Theorem 18.2 and Corollary 18.7 hold with $L+3$ in every length $L$. Hence $W(k,r) \le 2^{2^{r^{2^{2^{k+3}}}}}$ for every $k$, and Theorem 1.3 holds with density exponent $2^{-2^{k+3}}$.*

*The Bohr radius.* The threshold branch $64 \cdot 2^k$ of the Theorem 18.2 envelope comes from the inverse tight Bohr radius, which was bounded by comparison with the printed $\theta_1$ as $\exp(X^{2^{2^{k+6}+1}})$. Under the Fejér $\theta_1$ the radius is polynomial in $\theta_1'^{-1}$, so $\zeta^{-1} \le \exp(X^{2^{5k+21}})$. The threshold branch then drops to $16 \cdot 2^k + 5k + 25$, which is the threshold of Lemma 15.6. Every step costs at most $\max(\mu + 28 \cdot 2^k + O(k^2),\, 32 \cdot 2^k + 9)$.

**Theorem 10.** *Theorem 18.2 and Corollary 18.7 hold with $L+2$ at every length $L \ge 13$; in particular $W(k,r) \le 2^{2^{r^{2^{2^{k+2}}}}}$ and Theorem 1.3 holds with density exponent $2^{-2^{k+2}}$ for $k \ge 13$.*

*The base budget.* Every step also pays the base budget $\mu_0 \approx 16 \cdot 2^k$, and both of its sources are comparison bounds. Under the Fejér $\theta_1$ the fixed parameters (budget, $\theta_2/2$, $\delta$) are above $X^{-2^{5k+21}}$. The explicit family constants satisfy $K_k + p_k \le 2^{2^{4k+16}}$, a double logarithm linear in $k$, so $V \le 2^{2^{4k+13}}$. With $\mu_0 = 5k + 23$, one step costs only $12 \cdot 2^k + O(k^2)$ beyond the table, and both envelopes tend to $24 \cdot 2^k$.

**Theorem 11.** *Theorem 18.1 holds with exponent $\alpha^{2^{2^{d+3}}}$ at every degree $d \geq 11$, and with $\alpha^{2^{2^{d+4}}}$ at every degree $d \geq 8$. Theorem 18.2 and Corollary 18.7 hold with $L+1$ at every length $L \geq 13$, and with $L+2$ at every length $L \geq 10$. In particular $W(k,r) \leq 2^{2^{r^{2^{2^{k+1}}}}}$ and Theorem 1.3 holds with density exponent $2^{-2^{k+1}}$ for $k \geq 13$.*

*The class constant.* The split step only doubles the exponent: from $2^{k+6}a \leq 2^e$ and $D \leq 2^{e+s}$ it gives $2^{k+7}a' \leq 2^{2e}$ and $D' \leq 2^{2e+s+5k+22}$. So the invariant may start at any dimension from the exact values. At $l = 12$ they fit the constant $5/2$, and the class degree is then $\frac{5}{2}\,2^l + O(l^2)$ in every dimension beyond. Lemma 15.6's threshold, too, is polynomial under the Fejér $\theta_1$ ($X^{2^{5k+21}}$). Hence one step costs $7.5 \cdot 2^k + O(k^2)$, and both envelopes tend to $15 \cdot 2^k < 16 \cdot 2^k$.

**Theorem 12.** *Theorem 18.1 holds with exponent $\alpha^{2^{2^{d+2}}}$ at every degree $d \geq 17$. Theorem 18.2 and Corollary 18.7 hold with no offset, that is with $L$ in place of $L+9$ at the top of the threshold, at every length $L \geq 19$. In particular $W(k,r) \leq 2^{2^{r^{2^{2^k}}}}$ and Theorem 1.3 holds with density exponent $2^{-2^k}$ for $k \geq 19$.*

With the exact class table to dimension 14 the thresholds drop to $d \geq 16$ and $L \geq 18$ (and $k \geq 18$ in the consequences). Charging the partition constant of the step at $X^{2^{k+9}}$ instead of $X^{2^{4(k+3)^2+4}}$ lowers every Theorem 18.1 threshold by one more ($d+2$ from 15, $d+3$ from 10, $d+4$ from 7); in Theorem 18.2 the partition constant is an exponent of the refinement constant, so there it is a genuine cost.

Covering the two anchor slices of the single-piece lift jointly removes one class degree from every step. The lift applied the slice control twice, to $\varphi(\cdot, a)$ and then to $\varphi(\cdot, b)$ inside the first agreement set, at density $\theta_1\, c(c(\theta_1))$. Instead, both sections are covered at once at precision $\theta_1/4$ (Lemma 2). Then the density is $\theta_1(\theta_1/2)/Q(\theta_1/4)^2 = 8\, c(\theta_1/2)^2$, and the width is $\lceil L^{E(\theta_1/4)^2} \rceil$. Both step envelopes drop from about $15 \cdot 2^k$ to about $10 \cdot 2^k$. Theorem 18.1 then holds with $d+4$ in every degree, with $d+3$ from $d = 9$ and with $d+2$ from $d = 12$. Theorem 18.2, Corollary 18.7, the van der Waerden bound and Theorem 1.3 hold with $L+2$ from 9, $L+1$ from 12 and $L+0$ from 15. In Theorem 18.2 the refinement constant is raised to the power $P_{k+2} = ((k+2)!)^2\, 2^{(k+3)^2}$, so $\log_2 P_{k+2}$ is paid in every step. Since $(d!)^2\, 2^d \leq 2^{d^2}$, it is at most $(k+3)^2 + (k+2)(k+1)$, about half the bound $4(k+3)^2$ used before. This lowers the three Theorem 18.2 thresholds to $L+2$ from 8, $L+1$ from 11 and $L+0$ from 14. Finally, the split step bounds the count, the width exponent and the threshold of the next class separately and combined them by a sum; since every control is monotone in $(a, D)$, the maximum suffices, and the class degrees drop by 4–6%. This gives $d+2$ from $d = 11$. Keeping the count and the width exponent of a class on separate tracks lets dimension two start from its count $(n+1)^{11} W^{2^{41}}$ rather than the common $(n+1)^{32} W^{2^{48}}$; the class degrees drop by a further 6–16%, and $d+3$ holds from $d = 8$. The dimension-two count itself is $(n+1)^{11} W^{2^{39}}$ once Lemma 16.3's exponent 10002 is kept instead of being rounded to $2^{14}$, and with it Theorem 18.2, Corollary 18.7, the van der Waerden bound and Theorem 1.3 hold with $L+0$ from 13. This is the limit of the method: $d+1$ would need an envelope below $8 \cdot 2^k$, hence a class constant below 2, while the true constant is about 2.4.

## Summary of the constants

The best offsets proved, as of 2026-10-12, with the range of degrees $d$ or lengths $L$ (for $W(k,r)$ and Theorem 1.3, of $k$) in which each holds:

| | printed | $c$ everywhere | smaller $c$ from a threshold | | |
|---|---|---|---|---|---|
| Theorem 18.1, $\alpha^{2^{2^{d+c}}}$ | 10 | 4 | 3 ($d \geq 8$) | 2 ($d \geq 11$) | |
| Theorem 18.2, $L + c$ | 9 | 3 | 2 ($L \geq 8$) | 1 ($L \geq 11$) | 0 ($L \geq 13$) |
| Corollary 18.7, $L + c$ | 9 | 3 | 2 ($L \geq 8$) | 1 ($L \geq 11$) | 0 ($L \geq 13$) |
| $W(k,r) \leq 2^{2^{r^{2^{2^{k+c}}}}}$ | 9 | 3 | 2 ($k \geq 8$) | 1 ($k \geq 11$) | 0 ($k \geq 13$) |
| Theorem 1.3, $2^{-2^{k+c}}$ | 9 | 3 | 2 ($k \geq 8$) | 1 ($k \geq 11$) | 0 ($k \geq 13$) |

Each entry is a kernel-checked Lean theorem (§9). The thresholds come from dimensions 3 to 14, where the exact class degrees and the polynomial terms of each step dominate.

# 7 Theorem 16.2 and Corollary 16.11 with polynomial controls

The printed Theorem 16.2 restricts a relation with the product property to a set of density $1-\theta$ on which it is multiply linear, with graph count $q(r^{-1}\rho, \gamma, k)^r$ at precision $\rho$, where $r = \gamma^{-2} s(\theta, \gamma, k)$ and $s = (2/\theta\gamma)^{2^{2^{k+6}}}$. This count is exponential in a polynomial of $1/\theta$. The covers built for the dimension induction give a count that is *polynomial* in $1/(\gamma\theta\rho)$:

$$Q(\gamma, \theta, \rho) = \max\Big(1, \Big(\frac{4V}{\gamma\theta\rho}\Big)^{2^{68\cdot 2^k}}\Big), \qquad V = 2^{2^{2^{k+4}+1}},$$

with width exponent $Q^{-1}$, in every dimension $k \geq 1$ (`theorem_16_2_polynomial`). This form implies the printed one (`theorem_16_2_polynomial_implies_printed`).

The same holds for Corollary 16.11. For a function that is not uniform of degree $k+2$, the single-piece step with the joint-cover remainder gives a box of dimension $k+1$, of width $N^e$, on which a multilinear frequency is large on an $e$ fraction, with

$$e^{-1} \leq (2/\alpha)^{2^{236\cdot 2^k + k + 24}}$$

for every $k \geq 2$ (`corollary_16_11_polynomial`). The printed exponent is $(\alpha/8)\, c(r^{-1}\alpha/8, \alpha/2)^r$ with $r$ polynomial in $1/\alpha$, so its reciprocal is exponential in a polynomial of $1/\alpha$.

With the Fejér classes of §6 both exponents shrink:

- the graph count becomes $\max(1, (4V/\gamma\theta\rho)^{2^{23\cdot 2^k}})$ (`theorem_16_2_polynomial_fejer`);
- the corollary has $e^{-1} \leq (2/\alpha)^{2^{101\cdot 2^k + k + 24}}$ (`corollary_16_11_polynomial_fejer`).

With the dimension-two family exponent 32 (Theorem 8) they shrink further:

- the graph count becomes $\max(1, (4V/\gamma\theta\rho)^{2^{4\cdot 2^k + 3(k+4)^2 + 32}})$ (`theorem_16_2_polynomial_fejerS`);
- the corollary has $e^{-1} \leq (2/\alpha)^{2^{32\cdot 2^k + 9(k+4)^2 + k + 120}}$ (`corollary_16_11_polynomial_fejerS`).

With the class constant 5/2 and the polynomial base budget (Theorem 12) they shrink further, with $V = 2^{2^{4k+13}}$:

- the graph count has exponent $2^{t_k}$, where $t_k = \frac{5}{2}\,2^k + 3(k+4)^2 + 2$ for $k \geq 13$ and below that is the exact table (`theorem_16_2_polynomial_fejerX`);
- the corollary has $e^{-1} \leq (2/\alpha)^{2^{3t_k+6k+38}}$, that is $7.5 \cdot 2^k + O(k^2)$ (`corollary_16_11_polynomial_fejerX`).

With the exact class table to dimension 14, and with joint slices in the frequency box of the step, the frequency box costs one class degree less:

- the graph count has exponent $2^{t_k}$ with the exact table up to $k = 14$ (`theorem_16_2_polynomial_fejerZ`);
- the corollary has $e^{-1} \leq (2/\alpha)^{2^{2t_k+7k+38}}$, that is $5 \cdot 2^k + O(k^2)$ (`corollary_16_11_polynomial_fejerJ`).

At the two-track classes from the tight dimension-two count, the table $t_k$ is a further 6–22% lower in dimensions 3 to 14 (`theorem_16_2_polynomial_fejerT3`, `corollary_16_11_polynomial_fejerJT3`).

# 8 Further questions

1. *Joint slices.* The single-piece step applies its slice control twice in succession, $\theta\, c(c(\theta))$, and so pays $2 \cdot 68 \cdot 2^k$ in the step cost. Composing the two slice covers jointly, as in Lemma 2, should cost about $68 \cdot 2^k$. The envelope would drop from about $472 \cdot 2^k$ to about $336 \cdot 2^k$. This changes none of the constants in the table.
2. *Smaller offsets and thresholds.* Theorem 12 reaches $d + 2$ and $L + 0$, from degree 17 and length 19. Each step pays three class degrees, so a further offset needs a different step, for instance one that uses fewer vertex covers at full degree. The thresholds 17 and 19 come from dimensions 3 to 14. There the class table, 85 to 10375 in log-degree, and the polynomial terms $O(k^2)$ dominate.
3. *Fixed-precision slices with the Fejér classes.* Composing the two slice controls jointly would lower the step cost from $101 \cdot 2^k$ to about $78 \cdot 2^k$. That is still above $64 \cdot 2^k$, so it changes no constant in the table.

# 9 Formalization

All statements are in the directory `Combinatorics/Ramsey/Lean/GowersSzemeredi`.

| Statement | Lean |
|---|---|
| Lemma 2, Corollary 3 | `joint_cover_on`, `JointCoverOn.piece` (`Proofs16JointCovers`) |
| the remainder piece | `remainder_piece_of_vertex_covers` |
| the single-piece step | `single_piece_lift_global_cover` (`Proofs16SinglePieceCoverGlobal`) |
| $\rho_1$ bound | `rho1_inv_leC` (`Proofs18CoverDensity`) |
| envelopes | `polyGenMuC_le`, `genMuC_le`, `genMuC_le_sharp` |
| Theorems 18.1, 18.2, Cor. 18.7 | `theorem_18_1_explicit`, `theorem_18_2_explicit`, `corollary_18_7_explicit` (`Proofs18CoverExplicit`) |
| sharper constants | `Proofs18CoverSharper`, `Proofs18CoverTightBase`, `Proofs18CoverColorings`, `Proofs18CoverSharperExplicit`, `Proofs18CoverSharperLowDegrees`, `Proofs18CoverSharperShortLengths`, `Proofs18CoverThresholdsSharp`, `Proofs18CoverVanDerWaerden`, `Proofs18CoverHeadline`, `Proofs18CoverSharperAllDegrees` |
| Fejér Lemmas 15.5, 15.6 | `Proofs15FejerSelection`, `Proofs15FejerProduct` |
| tight radius, Fejér classes | `Proofs16TightRadius`, `Proofs16UniformCoverTwoFejer`, `Proofs16ClassStepDegreesFejer`, `Proofs16SplitStepFejer` |
| Fejér envelope | `Proofs18TightClassesFejer`, `Proofs18FejerEnvelopeBase`, `Proofs18CoverFejerHeadline` |

| Statement (continued) | Lean |
|---|---|
| Theorem 6 | theorem_18_1_d6, theorem_18_2_L4, corollary_18_7_L4, van_der_waerden_L4, theorem_1_3_L4 (Proofs18CoverFejerAllDegrees) |
| Theorem 7 | theorem_18_1_d5_twelve_le, theorem_18_2_L3_fourteen_le, Proofs18TightClassesFejerQ, Proofs18CoverFejerHeadlineQ |
| Theorem 8 | theorem_18_1_d4_thirteen_le, theorem_18_1_d5_nine_le, theorem_18_2_L3_eleven_le, theorem_1_3_L3_eleven_le, Proofs16UniformCoverTwoSplit, Proofs18TightClassesFejerS, Proofs18EtaStepSmall, Proofs18CoverFejerHeadlineS |
| Theorem 9 | theorem_18_1_d5, theorem_18_1_d4_eleven_le, theorem_18_2_L3, corollary_18_7_L3, van_der_waerden_L3_all, theorem_1_3_L3, Proofs18TightClassesFejerT, Proofs18CoverFejerHeadlineT, Proofs18CoverFejerLowDegrees5, Proofs18CoverFejerShortLengths3 |
| Theorem 10 | theorem_18_2_L2_thirteen_le, corollary_18_7_L2_thirteen_le, van_der_waerden_L2, theorem_1_3_L2_thirteen_le, zeta_gen_bounds_fejer5 (Proofs18CoverFejerHeadlineU) |
| Theorem 11 | theorem_18_1_d3_eleven_le, theorem_18_2_L1_thirteen_le, theorem_18_1_d4_eight_le, theorem_18_2_L2_ten_le, explicit_family_constants_poly_le, Proofs18FejerEnvelopeBaseW, Proofs18CoverFejerHeadlineW |
| Theorem 12 | theorem_18_1_d2_seventeen_le, theorem_18_2_L0_nineteen_le, corollary_18_7_L0_nineteen_le, van_der_waerden_L0, theorem_1_3_L0_nineteen_le, Proofs18TightClassesFejerX, Proofs18CoverFejerHeadlineX |
| joint slices | theorem_18_1_d4, theorem_18_1_d3_nine_le, theorem_18_1_d2_twelve_le, theorem_18_2_L0_fifteen_le, theorem_18_2_L1_twelve_le, theorem_18_2_L2_nine_le, Proofs16JointSliceCell, Proofs18JointSliceInverseFejer, Proofs18CoverFejerHeadlineJ, Proofs18CoverFejerLowDegrees4 |
| sharp partition constant | theorem_18_2_L0_fourteen_le, theorem_18_2_L1_eleven_le, theorem_18_2_L2_eight_le, polyPartConst_le_sharp (Proofs18CoverFejerHeadlineK) |
| split step by max | theorem_18_1_d2_eleven_le (Proofs18TightClassesFejerM, ...HeadlineJM) |
| two-track classes | theorem_18_1_d3_eight_le (Proofs16SplitTwoTrackClasses, ...HeadlineJT) |
| tight dimension two | theorem_18_2_L0_thirteen_le, theorem_1_3_L0_thirteen_le (...HeadlineK3) |
| 16.2 at $t_k$, 16.11 joint | theorem_16_2_polynomial_fejerZ, corollary_16_11_polynomial_fejerJ |
| Fejér polynomial 16.2, 16.11 | Proofs16PolynomialFejer, theorem_16_2_polynomial_fejer, corollary_16_11_polynomial_fejer |
| polynomial Theorem 16.2 | Proofs16PolynomialStructure, theorem_16_2_polynomial |
| polynomial Cor. 16.11 | Proofs16PolynomialCorollary, corollary_16_11_polynomial |

Every result above is kernel-checked. The production audit of 2026-10-11,

check_gowers.py GowersSzemeredi.CompletionAudit,

built 1,440 modules from source with Lean 4.32.0, checked the exact types of all 120 numbered catalogue statements and of the improved-bound theorems, and found only the axioms propext, Classical.choice and Quot.sound. The extensions to every degree ($c = 7$) and to every length ($c = 6$ and $c = 5$), the van der Waerden and Theorem 1.3 rows, and the polynomial Theorem 16.2 and Corollary 16.11 postdate that audit. They are so far checked against faithful stubs, and CompletionAudit.lean now includes them. The audit's record is

Research/GowersSzemeredi/verification/
2026-10-11-numbered-and-bounds.json.

The Fejér column (§6, 2026-10-12) is newer still. Its light modules are kernel-checked locally with the standard axioms only. Its bridge `Proofs18CoverFejerAllDegrees` is checked against a faithful stub whose axioms are exactly those of the $c = 7/c = 5$ bridge, namely the corpus theorems that the audit verified. It is not yet in `CompletionAudit.lean`. The research notes (`SECTION16_RESEARCH_NOTES.md`, §§L.11–L.16e) record the development.